Repurposing Geometric Foundation Models for Multi-view Diffusion

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Research Questions

- Can we leverage a latent space in which geometric structure is already encoded, rather than injecting or supervising it externally for novel view synthesis?
- What is the optimal boundary layer level for explicit synthesis when using multi-level features from geometric foundation models?
- How do geometric foundation model features compare to VAE/RAE latent spaces for multi-view diffusion in terms of training efficiency and geometric consistency?

Core Idea

Novel view synthesis requires geometric consistency across viewpoints, but existing diffusion methods operate in view-independent VAE latent spaces that lack geometric structure. The key insight is to repurpose the feature space of geometric foundation models (like Depth Anything 3) as the latent space for diffusion, since these models already encode strong cross-view geometric correspondences and 3D structure. This eliminates the need to learn geometry from scratch during diffusion training, leading to faster convergence and better 3D consistency.

Key Formula

Multi-level feature extraction

$$\{F_l\}_{l=0}^{L-1} = E_{geo}(I)$$

F_l is the multi-view feature sequence at level l with shape V×T×C, E_geo is the geometric foundation model encoder, I is the multi-view input images, L=4 levels. This extracts hierarchical geometric features that encode 3D structure across multiple viewpoints.

Joint source-target feature generation

$$\tilde{F}_l = [\tilde{F}_l^{src}, \tilde{F}_l^{tgt}]$$

F_l is the joint feature at level l, F^src_l are source features (N×T×C), F^tgt_l are target features (M×T×C). The diffusion model generates both source and target features jointly due to 3D attention coupling, then concatenates them along view dimension.

Key Figure

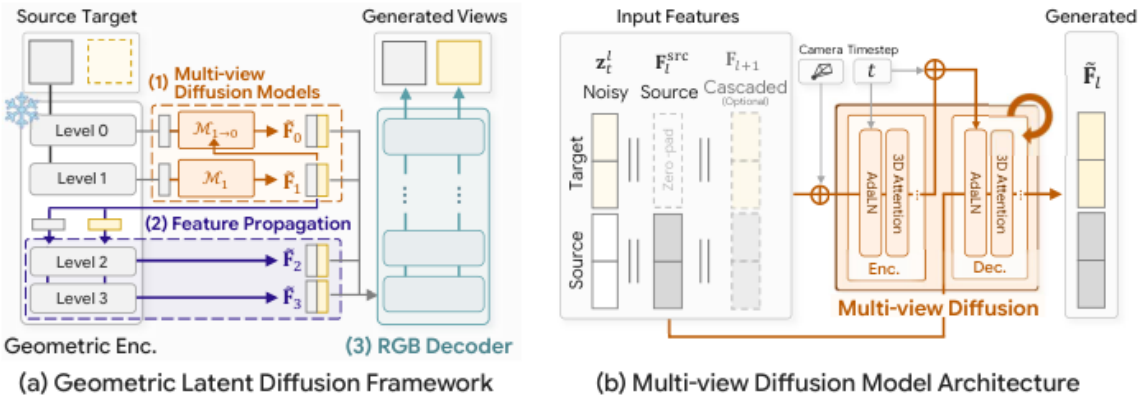

Fig. 2: Overview of the proposed framework. (a) Our NVS framework consists of three stages: **(1) multi-view diffusion models** synthesize features up to boundary layer k=1 via cascaded generation, **(2) feature propagation** derives deeper features through the frozen DA3 encoder, and **(3) the RGB decoder** maps the full multi-level features to target views. **(b)** The detailed architecture of a multi-view diffusion model.

Because the geometric foundation model's 3D attention jointly encodes source and target views, the resulting features are inherently coupled. We therefore generate both source and target features across all levels, denoted as $\{\tilde{\mathbf{F}}_l^{\text{src}}\}_{l=0}^{L-1}$ with $\tilde{\mathbf{F}}_l^{\text{src}} \in \mathbb{R}^{N\times T\times C}$, and $\{\tilde{\mathbf{F}}_l^{\text{tgt}}\}_{l=0}^{L-1}$ with $\tilde{\mathbf{F}}_l^{\text{tgt}} \in \mathbb{R}^{M\times T\times C}$. At each level l, the joint feature $\tilde{\mathbf{F}}_l \in \mathbb{R}^{(N+M)\times T\times C}$ is formed by concatenating the source and target features along the view dimension, yielding $\tilde{\mathbf{F}}_l = [\tilde{\mathbf{F}}_l^{\text{src}}, \tilde{\mathbf{F}}_l^{\text{tgt}}]$ with concatenation operator $[\cdot,\cdot]$. Finally, a dedicated RGB decoder $\mathcal{D}_{\text{rgb}}(\cdot)$ maps the complete synthesized feature set back to the pixel space to render the target views via $\tilde{\mathbf{I}}^{\text{tgt}} = \mathcal{D}_{\text{rgb}}(\{\tilde{\mathbf{F}}_l^{\text{tgt}}\}_{l=0}^{L-1})$. The target geometry $\tilde{\mathbf{G}}^{\text{tgt}}$ is also decoded such that $\tilde{\mathbf{G}}^{\text{tgt}} = \mathcal{D}_{\text{geo}}(\{\tilde{\mathbf{F}}_l^{\text{tgt}}\}_{l=0}^{L-1})$.

To this end, as illustrated in Fig. 2, **Geometric Latent Diffusion (GLD)** framework employs a three-stage pipeline. First, § 4.2 validates the reconstruction capacity of the geometric feature space by training $\mathcal{D}_{\text{rgb}}(\cdot)$ to decode multi-level features into RGB images. Second, to avoid the substantial cost of diffusing all L feature levels, § 4.3 identifies the optimal boundary layer k. Since deeper features can be obtained by propagating the boundary feature $\tilde{\mathbf{F}}_k$ through $\mathcal{E}_{\text{geo}}(\cdot)$, we only require explicit synthesis up to this boundary feature. Finally, § 4.4 employs a cascaded scheme to synthesize shallower features from $\tilde{\mathbf{F}}_k$ to ensure cross-level alignment.

4.2 Validating the Reconstruction Capability of Geometric Features

To validate the suitability of DA3's feature space for generative modeling, we first verify that its features can be decoded into high-fidelity images. We train a ViT-based decoder $\mathcal{D}_{\text{rgb}}(\cdot)$ to reconstruct RGB images from the multi-level features $\{\mathbf{F}_l\}_{l=0}^{L-1}$ extracted by the frozen encoder $\mathcal{E}_{\text{geo}}(\cdot)$. To ensure $\mathcal{D}_{\text{rgb}}(\cdot)$ effectively leverages the full signal, we introduce a level-wise dropout strategy during training. By randomly masking individual levels in $\{\mathbf{F}_l\}_{l=0}^{L-1}$, we force the decoder to reconstruct from partial inputs, improving its robustness.

Figure 2: Shows the complete GLD framework architecture with three stages: multi-view diffusion up to boundary layer k=1, feature

propagation through frozen encoder, and RGB decoding of full multi-level features.

Limitations

- Requires geometric foundation models as backbone, limiting flexibility to other architectures
- Computational overhead from multi-level feature processing and 3D attention mechanisms
- Performance depends on quality of pretrained geometric foundation model features
- Limited evaluation on out-of-domain scenes compared to methods with large-scale pretraining